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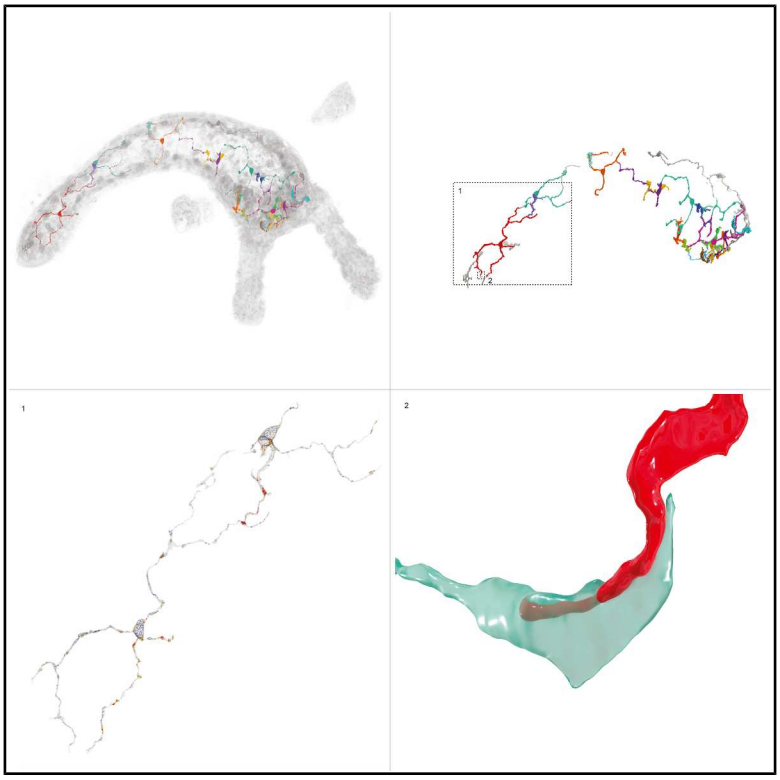
Ultrastructural reconstruction of the endodermal nerve net of *Hydra vulgaris*

Graphical abstract



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In brief

The cnidarian *Hydra vulgaris* has a simple nervous system yet is capable of complex behaviors. Zhang et al. reconstruct the ultrastructure of its endoderm nerve net, finding that most neuronal vesicles are located far from other neurons. Results suggest that *Hydra*'s nervous system operates via non-synaptic neuroendocrine transmission.

Highlights

- Five morphological neuronal subtypes present in *Hydra*'s endodermal nerve net
- Specialized intercellular junctions (handshakes) interconnect neurite tips of neurons
- Neuronal processes cross the mesoglea, connecting ectoderm and endoderm
- Most neuronal vesicles are located far from neighboring neurons

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Ultrastructural reconstruction of the endodermal nerve net of *Hydra vulgaris*

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SUMMARY

The cnidarian *Hydra vulgaris* has a simple nervous system with as little as a few hundred neurons distributed in two nerve nets in the ectoderm and endoderm. Using this simple neural chassis, *Hydra* can paradoxically perform relatively sophisticated behaviors, such as somersaulting and feeding. To understand how this simple nervous system is organized, we have performed a partial ultrastructural reconstruction of the endoderm nerve net of a small *Hydra* specimen. Neurons can be classified into 5 morphological subtypes, most of which connect via specialized interdigitations of their neurite tips, resembling handshakes. Neuronal processes cross the mesoglea, providing a means to coordinate the activity of the ectoderm and endoderm of the animal. Neurons have several different types of vesicles, including clear and dense core ones, which could support synaptic transmission. However, most vesicles are located far from other neurons. Moreover, neurite handshakes are mostly devoid of vesicles. We speculate that *Hydra*'s endodermal nerve net operates as a non-synaptic circuit, using a neuroendocrine chemical network to implement its functional operations.

INTRODUCTION

The cnidarian *Hydra vulgaris* is a freshwater polyp with a distributed nervous system without any cephalization or ganglia. Its neurons, from a few hundred to a few thousand,[1] depending on the size of the animal, belong to 11 transcriptomic cell types and are organized in 2 nerve nets,[2] in the ectoderm and the endoderm of the animal, separated by a mesoglea connective tissue.[3] Neurons activate two corresponding layers of epitheliomuscular cells, which control the movement of the hydroskeleton of the animal.[4] In spite of this simple structure, this nervous system enables *Hydra* to perform a surprisingly sophisticated repertoire of behaviors, from contraction, bending, and elongation to fixed action patterns such as feeding and somersaulting.[5] How such a simple nervous system can implement such a complex behavioral repertoire remains a fascinating mystery.[6]

Calcium imaging has enabled measurements of the activity of the entire nervous system of *Hydra*, revealing coactive neuronal ensembles associated with specific behaviors.[7,8] Neuronal ensembles in *Hydra* extend throughout the body of the animal, yet are non-overlapping, with neurons belonging exclusively to specific ensembles. This suggests that *Hydra*'s nervous system is built with separate circuits, each implementing a different function. To explore this, we performed ultrastructural reconstructions of *Hydra*'s endoderm, as it is smaller than the ectoderm. We reconstructed approximately half of the endoderm nerve net of one small specimen, which, given the radial symmetry of cnidarians,[9] should be representative of the rest of the endoderm. We characterized 20 complete neurons, also reconstructing 20 incomplete processes from neurons extending beyond our dataset. Endoderm neurons belong to morphological cell types that agree with the described transcriptomic ones. Interestingly, most neurons have long neurites connected at their tips with the tips of neurites of neighboring neurons, forming unusual interdigitating intercellular junctions ("handshakes"). Neuronal processes cross the mesoglea, thus connecting the ectoderm and endoderm. Finally, we detect several types of vesicles, including clear and dense core, typically involved in synaptic transmission. However, most vesicles, including fusing ones, are far from putative postsynaptic neuronal targets. Moreover, handshakes, which connect most neurons, are essentially devoid of vesicles. These findings, and the presence of a neuron that does not contact any other neuron, suggest that the main form of neural communication of *Hydra*'s endoderm nerve net is non-synaptic, using neuroendocrine release of transmitters or neuropeptides instead to form a chemical network.[10]

RESULTS

Reconstruction of neurons in *Hydra*'s endoderm

To study the structure of *Hydra*'s endoderm, we used a small animal (width 0.11 mm, total length 0.64 mm, body length 0.52 mm,

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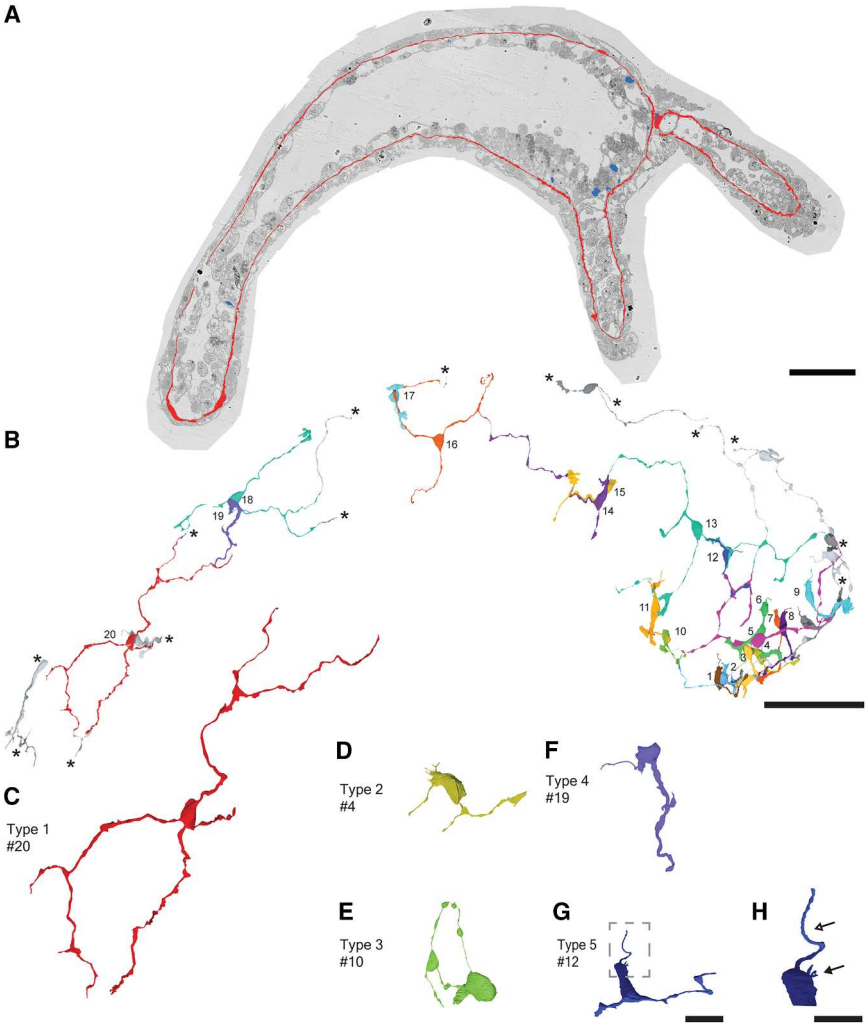

Figure 1. Partial reconstruction of *Hydra*'s endoderm nerve net

(A) Representative cross-section from scanning electron microscopy (SEM) image stack. The red line marks the mesoglea separating the endoderm and ectoderm, and the blue cells are endoderm neurons.

(B) 3D reconstruction of 20 endodermal neurons (randomly assigned colors). Stars mark 20 incomplete neurite ends (gray). Scale bar, 50 μm (A and B).

(C–G) Examples of type 1–5 neurons, as characterized in Table S1.

(H) Zoomed-in rendering of dashed box in (G), white arrow points to cilium and black arrow points to microvilli. Scale bar, 10 μm (C–G), 5 μm (H).

See also Figure S1 and Table S1.

average tentacle length 0.12 mm). Since neuron number in *Hydra* is proportional to body size,[11] a smaller animal contains fewer neurons while still maintaining essential morphological and functional properties of the nervous system. To preserve a near-native state, we used high-pressure freezing of the sample, followed by freeze-substitution fixation. Approximately half of the sample (0.05 mm thickness) was then cut along the oral-aboral plane into 1,829 thin sections (30 nm thick), collected and imaged with a scanning electron microscope at 4 nm resolution, and stitched and aligned to form image stacks (Figure 1A; see also Video S1). In this hemi-preparation, neurons were identified based on the presence of neurites, abundant dense-core and clear-core vesicles, and specialized structures such as handshakes, cilia, microvilli, and bulbous structures (described below). We reconstructed 20 complete neurons and 20 incomplete neuronal segments (Figure 1B). Reconstructed neurons were distributed throughout the endoderm with a higher density near the hypostome (see also Figure S1), with 11 neuronal cell bodies (44% of all neuronal cell bodies) located within a 65.04 μm^2 area (15% of the total body column area) on one representative section in the middle of the animal. Consistent with previous findings,[3,7] we found no neurons in the endoderm of tentacles.

Five major types of neuronal morphologies

Reconstructed neurons were classified into five morphology types (types 1–5) according to neurite polarity, contact (handshakes, further described below), and presence of cilia and microvilli (see also Table S1, ordered according to putative transcriptomic types[2]). Type 1 cells had handshake contacts and were either bipolar (#5 and 14) or multipolar (Figure 1C, #13, 16, 18, and 20). This morphology is consistent with previously described endoderm rhythmic potential 2 (RP2) cells, identified based on firing pattern,[7] and EN1 cells, based on RNA expression.[2] In both bipolar and multipolar cells, neurites extended horizontally, forming a planar lattice structure, like a net, consistent with the EN1/N4 neuron morphology from transgenic animals,[2,12] and contacted endoderm epithelio-muscular cells. Additionally, we identified two neuronal classes, types 2 (Figure 1D, #2 and 4) and 3 (Figure 1E, #10), which were bipolar and also had handshake contacts as type 1. However, type 2 neurons had cilia and microvilli, both of which were absent in type 1. Type 2 neurons were located near the hypostome and may correspond to sensory neurons.[12] The only type 3 neuron had a peculiar set of semi-circular neurites connected by a handshake. In addition, we found four unipolar neurons (#7, 8, 9, and 19) with one neurite, which we classified into type 4, and seven pseudo-unipolar neurons (#1, 3, 6, 11, 12, 15, and 17) with one short neurite that split into two branches, classified into the type 5 class. Type 4 cells (Figure 1F, #7 and 19) may correspond with EN2 cells,[2] whereas pseudo-unipolar type 5 neurons (Figure 1G, #1, 3, 6, 11, 12, 15, and 17) may correspond with EN3 cells. Type 5 cells had microvilli surrounding the main cilium that were absent in type 4 (Figure 1H).

Neuronal morphological features and contacts

We noticed several interesting morphological features in reconstructed neurons. Every type 2, 4, and 5 neuron (except for #7)

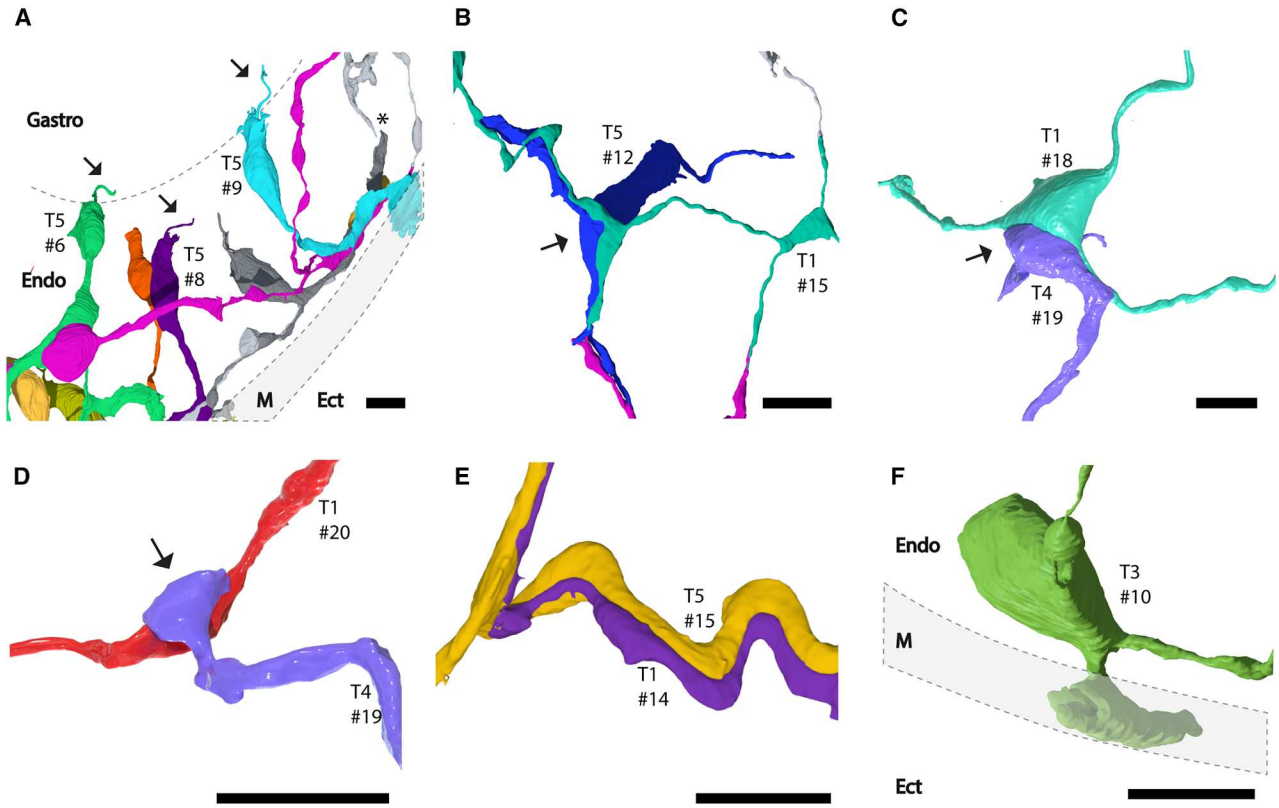

Figure 2. Neuronal features

(A) Type 5 neurons in endoderm (Endo) had cilia (arrows) on their apical pole in gastrovascular cavity (Gastro). Mesoglea (M, in gray) and ectoderm (Ect) outlined by a dotted line.

(B) Neurite (type 1, neuron #15) to soma (type 5, neuron #12) contact (arrow).

(C) Soma (type 1, neuron #18) to soma (type 4, neuron #19) contact (arrow).

(D) Neurite (type 4, neuron #19) to neurite contact (type 1, neuron #20) marked by an arrow.

(E) Neurites (type 1, neuron #14, and type 5, neuron #15) in parallel contact.

(F) Bulbous structure on soma (type 3, neuron #10) ended in the mesoglea. Scale bar, 5 μm (A–F).

had one cilium (107.93 ± 1.25 μm, mean ± SEM, n = 12) and multiple microvilli (except for type 4, 8.25 ± 1.63, mean ± SEM, n = 12) that extended into the gastrovascular cavity (Figure 2A). This is consistent with the previous finding[13] that neurons in the pedal disk of the endoderm lack cilia and suggests a role of cilia in receiving sensory signals from the gastric cavity. In three type 5 neurons (#12, 15, and 17), the somata were attached to a neurite of type 1 neurons (Figure 2B). Also, all type 4 neurons (#7 and 19) were attached to the soma (Figure 2C) of another neuron (types 5 and 1). One type 4 neuron (#19) also had a neurite that ended on the neurite of a type 1 (#20) (Figure 2D). Eight type 5 neurons (#8, 9, 1, 3, 6, 12, 15, and 17) formed parallel contacts with type 1 or 2 neurons (Figure 2E). We also found large bulbous structures in all neuron types (width 3.42 ± 0.37 μm, length 6.51 ± 0.61 μm, mean ± SEM, n = 13) that emerged from the soma or neurites and were embedded in the mesoglea (Figure 2F).

Specialized handshake contacts between neurons

We found a peculiar structure at the end of the neurites of all type 1, 2, and 3 cells, whereby the ends of the neurites were physically interdigitated, like a handshake, with one neurite entering the neurite from another neuron and becoming engulfed by it (Figure 3A). Interestingly, we did not observe ultrastructural evidence of chemical synapses at these locations (Figures 3A1 and 3A2). Engulfed segments were 0.77 ± 0.19 μm in length and had portions of 1.03 ± 0.20 μm of parallel, un-engulfed segments (mean ± SEM, n = 12). In 9 neurons, both neurites entered each other and were mutually engulfed by the other neuron, as a "two-way handshake" (Figures 3B, 3B1, and 3B2). In these cases, engulfed segments were 0.70 ± 0.18 μm in length with 2.22 ± 0.54 μm of external segments (mean ± SEM, n = 12). In one type 3 neuron (#10), a handshake was formed between two of its own neurites, potentially creating a ring neurite (Figures 3C and 3C1). However, there was an additional mitochondria-containing segment without a clear soma that ran parallel to the handshake (Figure 3C2), making it difficult to confirm that the loop was indeed connecting neurites from the same neuron.

Neurites extend through the mesoglea

Previous ultrastructural studies described that no neural processes cross the mesoglea.[3] This means that the two nerve nets are not connected, ruling out both electrical and chemical

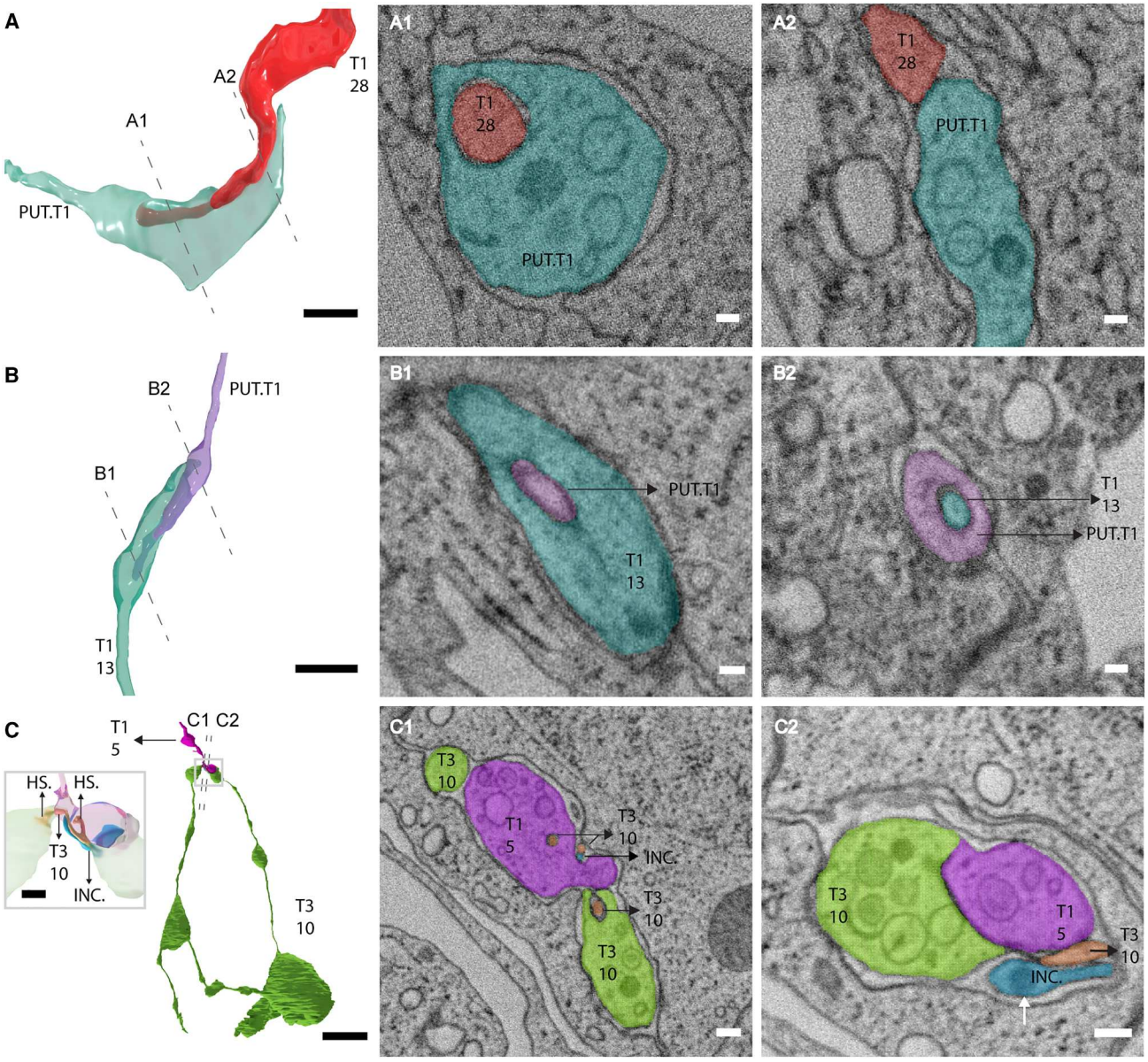

Figure 3. Neurite handshake contacts

(A) One-way handshake where a type 1 neuron (#28, red) enters a putative type 1 incomplete neurite (PUT.T1, blue). Scale bar, 1 μm. (A1) Cross-section EM image at (A1) in (A). (A2) Cross-section EM image at (A2).

(B) Two-way handshake where type 1 incomplete neurite (PUT.T1, purple) and type 1 neuron (#13, blue) neurite enter and end inside each other. Scale bar, 1 μm. (B1) Cross-section EM image at (B1) in (B) of a purple neurite engulfed by a blue neurite. (B2) Cross-section EM image at (B2) of blue neurite engulfed by purple neurite. Scale bar, 100 nm (A1, A2, B1, and B2).

(C) Type 3 neuron (#10) with interconnected neurites that formed a circle. Scale bar, 5 μm. The gray box is a zoomed-in rendering of the handshake of one neurite of neuron #10 (in yellow) entering the green neurite. A second handshake is shown, with a yellow neurite entering a magenta neurite (type 1, neuron #5) and a short segment of an incomplete cell (blue). Scale bar, 500 nm. (C1) Cross-section EM image at (C1) in (C), showing the yellow neurite (type 3, neuron #10) entering the green neurite (type 3, neuron #10) and the magenta neurite (type 1, neuron #5) forming two handshakes. (C2) Cross-section EM image at (C2) in (C), showing an incomplete cell segment (INC, blue) that had mitochondria (Mito) in the cytoplasm but did not have a connected soma. Scale bar, 200 nm (C1 and C2).

synapses, and raising the conundrum of how the two different parts of the nervous system in *Hydra* can communicate with each other. We re-examined this issue in our dataset. Although most endoderm neurons had processes located only on the endoderm, we found neurites crossing the mesoglea and reaching the ectoderm. Specifically, two endoderm neurons in the hypostome (type 2, #4 and type 5, #3) crossed the mesoglea and made contact with six ectoderm neurons (Figures 4A and 4B). Three neurons in the hypostome (type 2, #2; type 5, #6, and incomplete), two neurons in the body column (type 5, neuron #15, and incomplete), and one neuron near the foot (incomplete) also crossed the mesoglea and made contact

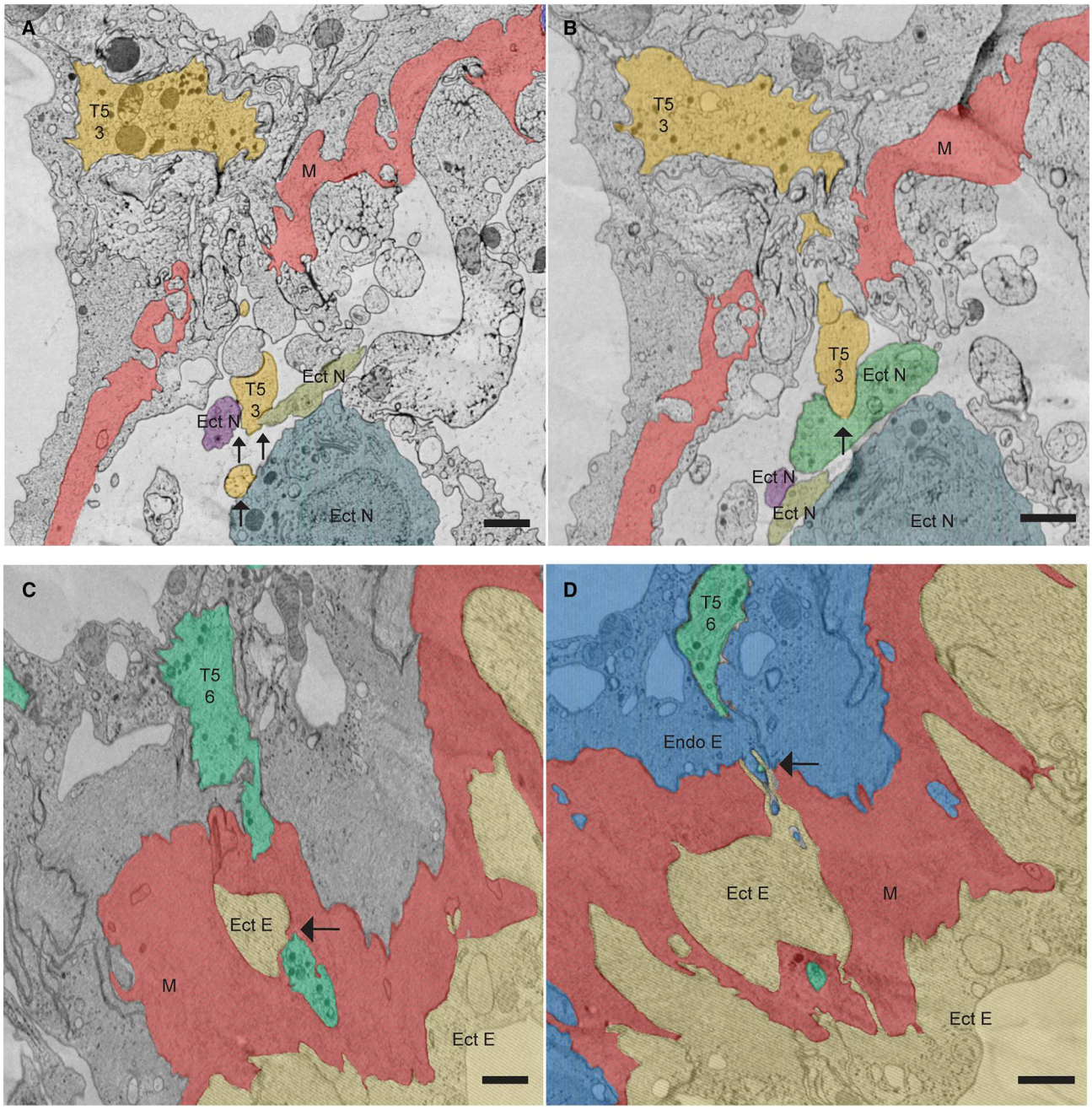

Figure 4. Neurites crossing the mesoglea

(A and B) Two sections, 570 nm apart, showing one endoderm neuron (type 5, #3, orange) crossing mesoglea (M, red) and contacting (arrows) three ectoderm neurons (Ect N in purple, yellow, dark green, and dark blue) in (A), and one ectoderm neuron (green) in (B).

(C) Endoderm neuron (type 5, #6, green) to ectoderm epithelial cell (Ect E, yellow) contact (arrow).

(D) Endoderm epithelial cell (Endo E, blue) to ectoderm epithelial cell contact (arrow). (C) and (D) are 13 slices, 390 nm apart. Scale bar, 1 μm (A–D).

with ectoderm epithelial cells (Figure 4C). Inside the mesoglea, in addition to confirming previously described contacts between endoderm and ectoderm epitheliomuscular cells[3,12,14] (Figure 4D), we also observed seven endoderm neurons (type 1, #14; type 5, #11, 17, and 4 incomplete neurites) that contacted ectoderm epitheliomuscular cells through bulbous structures embedded in it.

Endoderm neurons have different classes of vesicles

We studied the vesicles present in endoderm neurons with an algorithm to detect them automatically (see STAR Methods). 53,851 vesicles were identified in the 20 complete neurons. We classified vesicles into five subtypes, following previously described vesicle types in *Hydra*.[15,16] The first type was dense-core vesicles (DCVs; Figure 5A), with uniformly dense centers, a

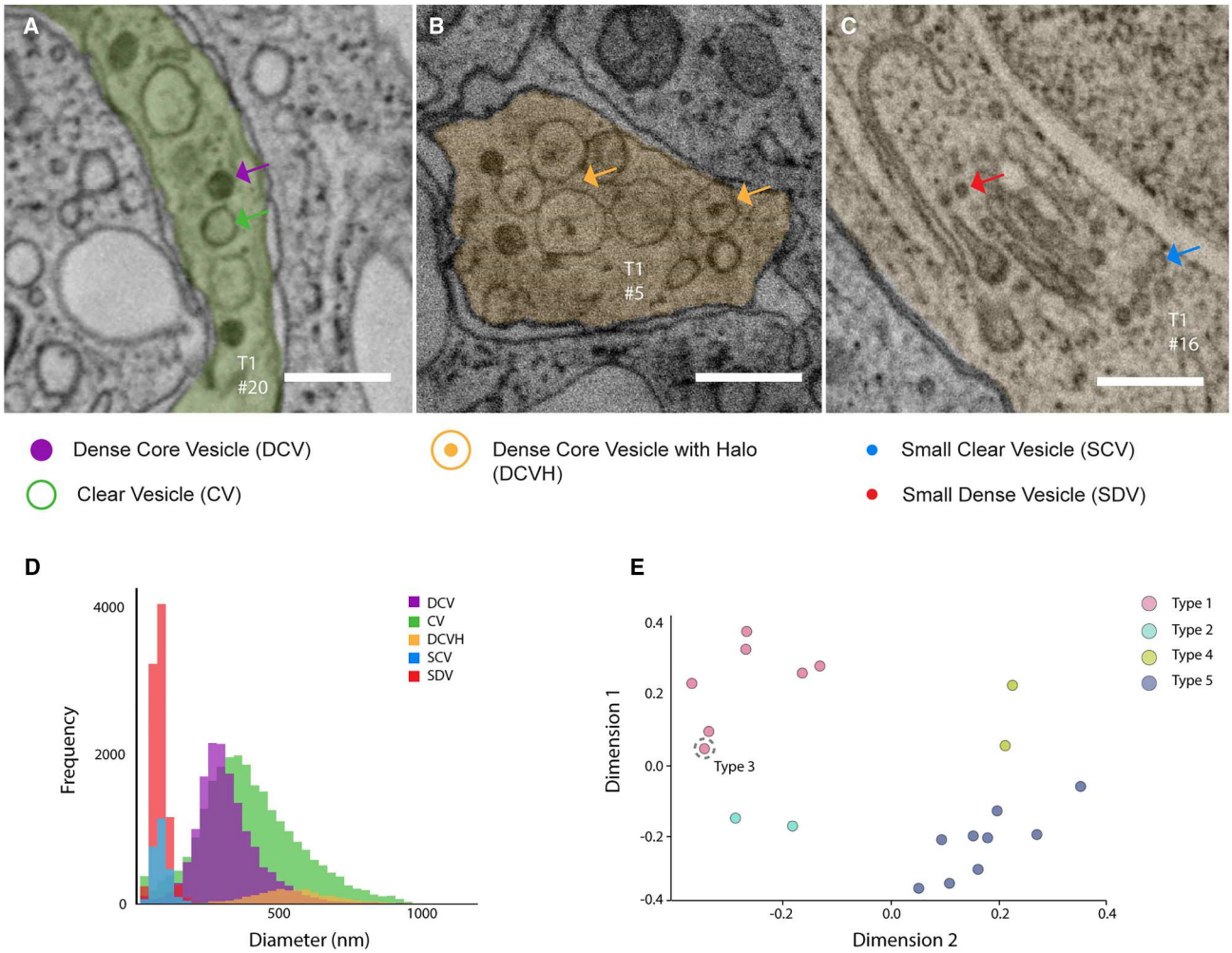

Figure 5. Types of neuronal vesicles

(A) Example of dense core vesicles (DCVs; purple arrow) and clear vesicles (CVs; green arrow) in type 1 neurite (#20).

(B) Example of DCVs with halo (DCVH; orange arrow) in type 1 neurite (#5).

(C) Example of small clear vesicles (SCVs; red arrow) and small dense vesicles (SDVs; blue arrow) in type 1 soma (#16). Scale bar, 500 nm (A–C).

(D) Diameters of different types of vesicles: DCV (purple), CV (green), DCVH (orange), SCV (blue), and SDV (red).

(E) Neuronal classification based on vesicle types and morphology features. Clustering using Gower's distance and MDS, with neurons color coded by type. See also Figure S2.

volume of 2,673,934.62 ± 18,317.32 nm^3, and a diameter of 316.69 ± 0.93 nm (extrapolated by collapsing spherical vesicles to circles and converting area to diameter assuming each vesicle is a perfect sphere; n = 15,188). The second type were clear vesicles (CVs), with clear or slightly dense centers (4,492,699.00 ± 25,720.50 nm^3 in volume and 398.36 ± 1.13 nm in diameter, n = 24,998). A third type were DCVs with a halo (DCVH), with a dense center and a light halo around them (Figure 5B; 7,495,975.17 ± 81,531.69 nm^3 in volume and 546.19 ± 3.11 nm in diameter, n = 2,049). We also found smaller vesicles, which we operationally defined as having a diameter smaller than 80 nm (n = 11,616), and further classified into two additional types: small clear vesicles (SCVs), with a clear center, and small dense vesicles (SDVs), with a dense center (Figure 5C). Small vesicles were mostly found in the soma (88.92% ± 1.44%, n = 20), near the Golgi apparatus or in the process of budding off. These five types of vesicles had a multimodal distribution of diameters, consistent with the possibility that they represent five different subtypes (Figure 5D).

Hierarchical clustering of vesicle types confirms neuronal morphological classification

We made use of the detailed information on vesicle types and the numbers of vesicles per neuron to quantitatively reexamine our initial classifications of neurons into 5 morphological types, based only on qualitative criteria. For this, we quantified all the used morphology features (polarity, cilia, microvilli, and handshakes), neuronal measurements (total volume, nucleus volume, and total length), and vesicle type composition (percentage of each type of vesicle) of each neuron and used Gower's distance matrix and multidimensional scaling (MDS) to explore the possible existence of neuronal subtypes (Figure 5E). These results revealed 4 clusters at a distance threshold of 0.4

(see also Figure S2), which essentially aligned with the five morphological subtype clusters from our initial classification, with the exception of type 3 neuron #10, which was classified as type 1. These quantitative results, blind to the initial subgroups, validated the classification of endodermal neurons and indicated the similarity between type 1 and 3 neurons.

Docked and fusing vesicles mostly apposed to non-neuronal cells

We found it difficult to identify clear synaptic profiles, as traditionally defined by vesicles that are docked or present in large numbers across a potential synaptic target cell with a thicker postsynaptic membrane and a wider synaptic cleft. This difficulty could be due to the limited resolution of the scanning electron microscope, as compared with transmission electron microscopy (TEM), or to the possibility that synapses in cnidarians are morphologically distinct from those in bilaterians.[15] To further explore the presence of synapses, we searched for evidence of potential vesicle release sites as a telltale sign of a synaptic contact. Most vesicles in contact with the plasma membrane were apposed to non-neuronal cells, including 5,896 docked vesicles (Figures 6A–6C) and 108 fusing vesicles, forming omega shapes (Figure 6D). Only 527 docked vesicles (Figures 6E–6G) and 4 omega-shaped vesicles (Figure 6H) were apposed to other neurons. Statistical analysis confirmed the preference for docked (Figure 6I) and omega-shaped vesicles (Figure 6J) to be apposed to non-neuronal targets (including epitheliomuscular cells, gland cells, and within mesoglea) as opposed to other neurons, across all neuronal cell types (except for type 4, which only had 2 data points; see also Table S2). For this analysis, types 1, 2, and 3 were grouped together because they all had handshake structures and are potentially EN1 neurons.

Most neuronal vesicles are located far from neuronal contacts

The previous results suggested that most neuronal vesicles were located near non-neuronal cells. To further explore this, we analyzed the location of different types of neuronal vesicles and visually rendered vesicles in the complete neurons and overlaid them with a translucent neuron mask, where vesicles were color-coded by type (Figure 7A; see also Figure S3; Extended Data 1). Vesicles were distributed throughout the neuronal cytoplasm but generally far from locations of neuronal contacts. Density maps of vesicles consistently showed high density of vesicles in the somata and areas of the neurites that were far from other neurons (see also Figure S4A; Extended Data 2). We also found a high density of large vesicles on neurites near neighboring non-neuronal cells (Figure 7B; see also Figure S4B; Extended Data 3). The increased proportion of smaller vesicles in the soma suggests a relation with the Golgi apparatus.

Quantification revealed that most large vesicles (68%–81%) were located further than 708 nm (a distance corresponding to the mean plus 2 standard deviation of all large vesicle diameters) from the closest membrane of another neuron (Figure 7C; see also Table S3). This was consistent across large vesicle types and across all neuron types (Figures S5A–S5C; see also Table S3). Since close to half (45%) of neurons belonged to types 1, 2, and 3, we focused on them and determined what percentage of their vesicles were located close to other type 1, 2, and 3 neurons. We found that practically all (97.97% ± 0.27%, mean ± SEM, n = 9) large vesicles (clear or dense core) in handshakes were located at least 708 nm away from the membrane of another neuron of their class. Due to the lack of clear synaptic structures in handshakes, we were unable to determine the pre- and post-synaptic partners; however, the number of vesicles on either side of the handshake contacts was also small (17.57 ± 2.86, mean ± SEM, n = 21; see also Table S4). These analyses confirmed that only a paucity of vesicles was in locations where they could be involved in neuron-to-neuron synaptic transmission.

Isolated neuron containing many vesicles in a spherical protrusion

Notably, we found one type 5 neuron (#11), which was not in contact with any other neuron (Figure 7D), yet had one cilium and four microvilli extended into the gastrovascular cavity and a spherical process adjacent to a gland cell (Figure 7E; see also Figure S5D). This neuron contained a large number (3,893) of vesicles, including 1,298 DCVs, 794 CVs, 19 DCVHs, 1,005 SDVs, and 777 SCVs (Figure 7F). The spherical process at the end contained a high density of large vesicles (54.28% of total large vesicles in 5.19% of neuronal volume) with a high concentration of DCVs (79.49% in the sphere, 40.10% in the rest of the neuron) and a lower concentration of CVs (20.42% in the sphere and 58.03% in the rest of the neuron). As the neuron did not contact any other neurons, these vesicles could only be released in a non-synaptic mechanism.

DISCUSSION

Different morphological types of neurons in *Hydra*'s endoderm

In this study, we performed a partial ultrastructural reconstruction of the endoderm nerve net of a small *Hydra* specimen, distinguishing five different subtypes of neurons (Table S1). Quantitative analysis confirmed the existence of four of these five subtypes, with two of them (types 1 and 3) possibly representing the same population (Figure 5E). These neuronal subtypes appear in good agreement with described transcriptomic cell types described in the *Hydra* endoderm, assuming that the three subtypes that we identified (types 1–3) correspond to transcriptomic subtype EN1.[2] In our dataset, different subtypes of neurons were located throughout the endoderm of the animal, except for the tentacles.[3,7] We also found a higher neuronal density near the hypostome, most of them with a cilium. This subnetwork could represent a sensory hub of a stomatogastric nervous system.[17]

Interdigitating junctions between neurons

An unexpected finding was a highly specialized morphological structure that connects the tips of the dendrites of all type 1–3 neurons. Interdigitating processes (handshakes) were generated by protrusions from the tip of a neurite of one neuron entering the tip of the dendrite of another (Figure 3). Handshakes were the only point of contact between type 1–3 neurons, which were also the majority of the neurons and the ones with the most extensive neurites. Thus, the handshakes can be viewed as the major connecting points for the endoderm nerve net. Since they lack synaptic profiles, they could

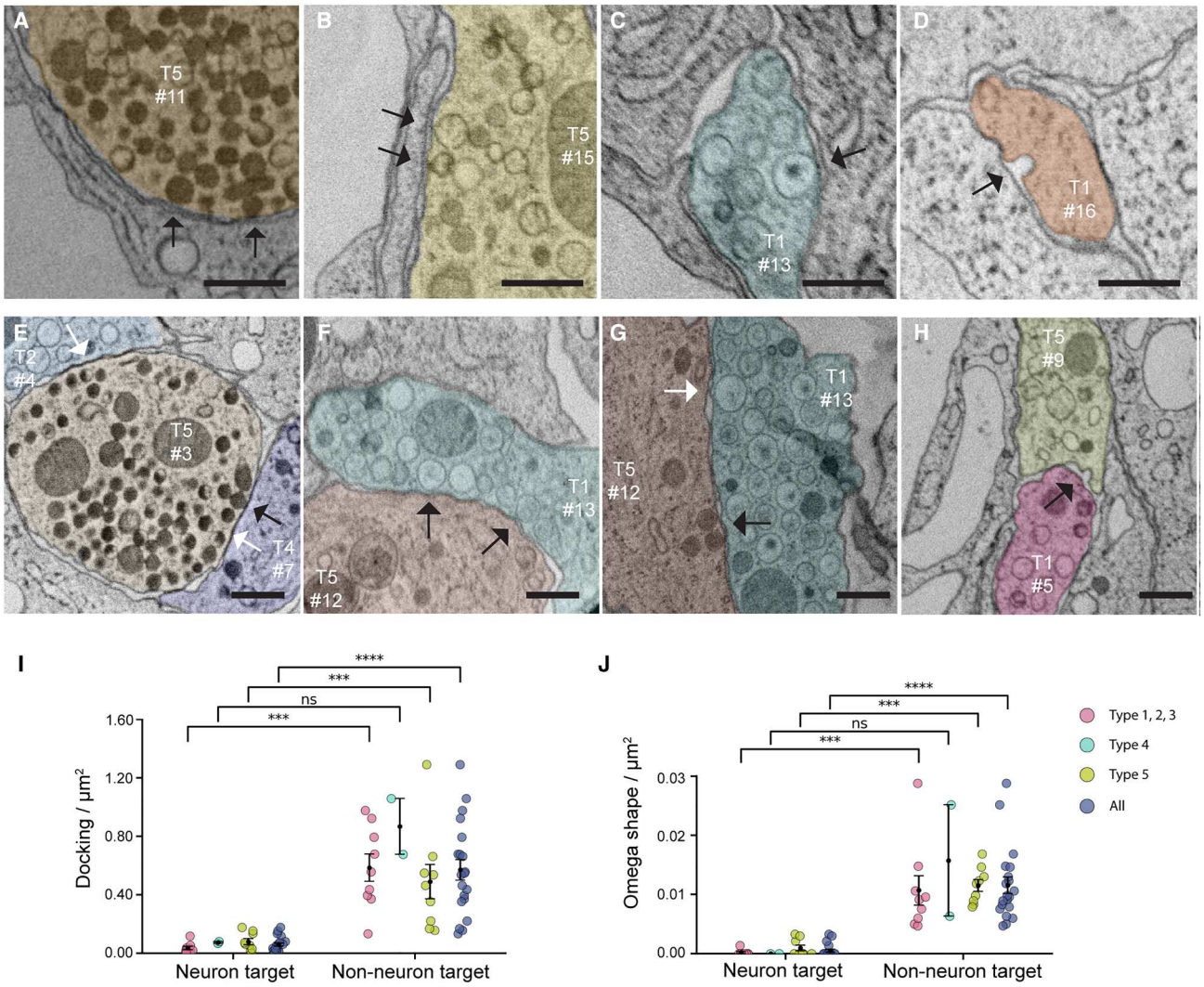

Figure 6. Docked and omega-shaped vesicles

(A) Two docked DCVs (arrows) at the membrane of a type 5 neuron, apposed to an epitheliomuscular cell.

(B) Two docked CVs (arrows) in a type 5 neuron apposed an epitheliomuscular cell.

(C) Docked DCVH (arrow) in a type 1 neuron near an epitheliomuscular cell.

(D) Omega-shaped vesicle (arrow) in a type 1 neuron near an epitheliomuscular cell.

(E) One docked DCV (black arrow) at the membrane of a type 5 neuron near a type 4 neuron. Potential gap junction between type 5, 2, and 4 neurons (white arrows).

(F) Docked CVs (arrows) in a type 2 neuron near a type 5 neuron.

(G) Two docked DCVs (black arrows) in a type 5 neuron near a type 1 neuron. A reciprocal docked CV (white arrow) at the membrane of a type 1 neuron near a type 5 neuron.

(H) Omega-shaped vesicle (black arrow) in a type 5 neuron apposed to a type 1 neuron. Scale bar, 500 nm (A–H).

(I) Docked vesicles near neuron target and non-neuron target, normalized by surface area in type 1, 2, and 3 neurons; type 4 neurons; type 5 neurons; and all neurons (Mann-Whitney U test). Mean ± SEM, ns $p > 0.05$, ***$p < 0.001$, ****$p < 0.0001$; types 1, 2, and 3, $n = 9$; type 4, $n = 2$; type 5, $n = 9$; all, $n = 20$; Mann-Whitney U test.

(J) Omega-shaped vesicles near neuron targets and non-neuron targets normalized by surface area in type 1, 2, and 3 neurons; type 4 neurons; type 5 neurons; and all neurons (Mann-Whitney U test). Mean ± SEM, ns $p > 0.05$, ***$p < 0.001$, ****$p < 0.0001$; types 1, 2, and 3, $n = 9$; type 4, $n = 2$; type 5, $n = 9$; all, $n = 20$; Mann-Whitney U test.

See also Tables S2 and S3.

accommodate gap junctions to effectively connect all of the type 1–3 neurons into a single synchronized network.[18,19] Consistent with this, calcium imaging of endoderm neurons reveals the presence of a synchronized ensemble (RP2), most likely due to the type 1–3 EN1 subtype of neurons.[7] Another possibility is that handshakes could provide anchors to physically stabilize the endoderm nerve net through maximized membrane apposition. Handshakes could also play a role in tension sensing, potentially contributing to water homeostasis and the regulation of body contractions.

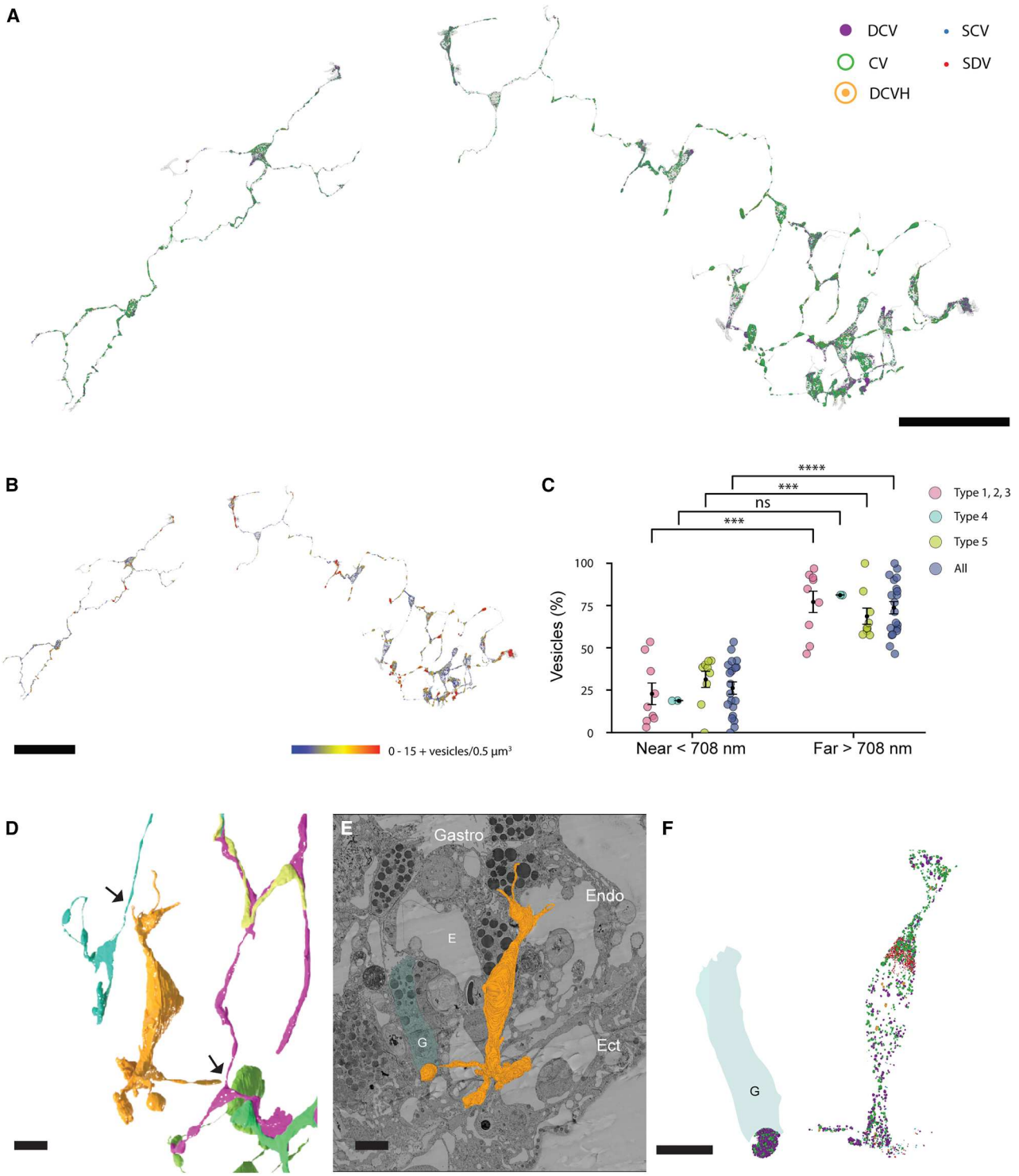

Figure 7. Vesicles are far from other neurons

(A) Location of all vesicles in neurons: CV (green), DV (purple), DCVH (orange), SCV (blue), and SDV (red). Scale bar, 50 μm. See also Figure S3 and Extended Data 1.

(B) Density of large vesicles in neurons, defined as the number of vesicles located within a 0.5 μm^3: 0: blue, 15+: red (Extended Data 3).

(C) Percentage of large vesicles in neuronal subtypes within or beyond 708 nm from another neuronal membrane (Table S3). Mean ± SEM, ns $p > 0.05$, ***$p < 0.001$, ****$p < 0.0001$; types 1, 2, and 3, $n = 9$; type 4, $n = 2$; type 5, $n = 9$; all, $n = 20$; Mann-Whitney U test.

(legend continued on next page)

Endoderm and ectoderm crosstalk

Previous results have documented the absence of neuronal processes crossing the mesoglea in *Hydra*,[3] making it difficult to explain how the ectoderm and endoderm work synergistically to generate coordinated neuronal activity,[8] muscle activity,[4] and behaviors.[5] However, we found several neurites in the mesoglea, providing direct communication between endoderm and ectoderm neurons and between endoderm neurons and ectoderm epitheliomuscular cells. This could explain the coordinated neuronal and muscle activity between both tissues. In addition, ectoderm and endoderm neurons could also be functionally interconnected due to a chemical network of neuropeptides (see below).[8] Finally, the mechanical coupling of ectoderm and endoderm to the hydrostatic skeleton of the animal[20] could indirectly provide another mechanism for functional communication between the two nerve nets of the *Hydra*.

Different morphological types of vesicles

Our study was carried out to map the synaptic circuits underlying the synchronized neuronal ensembles found with calcium imaging,[7] but we found it difficult to identify synapses. However, we found over 53,000 vesicles that belonged to a variety of morphological types, consistent with previous descriptions of neuronal vesicles in cnidarians and other species.[15,16,21–24] Specifically, we found large CVs, interpreted to contain small amino acid neurotransmitters like glutamate and γ-aminobutyric acid (GABA), and large dense or dense core with halo vesicles, which normally carry neuropeptides[25] (Figure 5). We also found smaller CVs, particularly in the somatic compartment, potentially recycling endosomes or parts of the Golgi vesicular apparatus. Every neuron contained all types of vesicles, in principle inconsistent with Dale's principle that each neuron expresses only one type of neurotransmitter.[26] At the same time, different types of neurons had a different complement of vesicle types, which helped to quantitatively differentiate neuronal subtypes (Figure 5E). Although we did not identify clear synaptic profiles, we found vesicles in the process of fusing with the membrane (Figure 6). Omega shapes, whereby vesicles are incompletely fused, are interpreted as morphological evidence of transmitter release in other species.[27] However, most profiles of fusing vesicles occurred apposed to the membranes of non-neuronal cells, generally epithelium-muscular cells, consistent with neuromuscular transmission.

Vesicles located far from neuronal contacts

A salient finding was that only a small proportion of vesicles were near membranes of the closest neuron, its putative synaptic partner (Figure 7), and even these low numbers could be an overestimate, as most vesicles that were close to other neurons were in neurites that fasciculated together but may not be functionally interacting. Moreover, in type 1–3 neurons, which had the most extensive arborizations and formed the handshakes, large vesicles were practically excluded from contact points with other neurons of the same type. If chemical synapses were the major mechanism of neuronal communication in the endoderm nerve net of *Hydra*, one would have expected to find synaptic vesicles in close apposition to their potential neuronal targets. In addition, the presence of one isolated neuron, loaded with vesicles yet without any physical contact with other neurons (Figures 7D–7F), demonstrates that, at least for this neuron, neuronal signaling cannot be synaptically mediated. This isolated neuron resembled mammalian chromaffin cells and may modulate digestive processes, as it had cilia that could sense the internal environment of the gastric cavity and a knob of vesicles apposed to secretory gland cells that could release enzymes into it.

A neuroendocrine chemical network in *Hydra*'s endoderm?

The lack of clear synaptic profiles and the location of neuronal vesicles far from other neurons in our data are consistent with the possibility that the endoderm nerve net of *Hydra* operates as an essentially chemical network, with paracrine release of neurotransmitters and neuropeptides and without using chemical synapses. The possibility that neurons communicate via non-synaptic chemical signals has been raised before, from Paul Weiss' early hypothesis of a chemical "radio broadcast" as the fundamental mode of communication in the nervous system[28] to the discoveries of a large number of neuropeptides and receptors in cnidarians.[29–32] In other non-bilaterians, ultrastructural reconstructions demonstrate neurons that are physically isolated from others and have few synaptic profiles. In fact, the ultrastructure evidence of chemical synapses in non-bilaterian and early invertebrates is not strong and could reflect a difference in the balance between synaptic and non-synaptic circuits when compared with bilaterians that have centralized nervous systems and specialized synaptic architecture.[33]

A chemical "wireless connectome" seems well suited for a small cnidarian such as *Hydra*, whose nerve net is organized as a planar sheet of neurons embedded in an epithelial layer. This may allow chemical signals to diffuse efficiently across the body, as compared with larger, multilayered nervous systems. Moreover, a chemical signaling system could be resilient to major body deformations that occur during *Hydra*'s behavior, something that could be damaging to a nervous system based on synaptic hardwiring. A chemical connectome could also be easier to scale up or down than a synaptic connectome, adapting well to *Hydra*'s remarkable order of magnitude changes in size, depending on its nutritional state. A neuroendocrine network could also facilitate *Hydra*'s remarkable regenerative capabilities, including the continuous displacement of tissue, the continuous insertion of new neurons into the existing nervous net, and even the reaggregation of dissociated cells into a complete animal,[34] since the maintenance and rewiring of the nervous system could be more difficult if it used chemical synapses.

(D) Reconstruction of isolated neuron without neuronal contacts. Arrows point at locations where other neurons were nearby, but without contacts.
(E) Section of reconstruction of same neuron showing adjacent cells surrounding the neuron, including gland (G) and epithelial cells (E). Note cilia extended into the gastrovascular cavity (Gastro).
(F) Vesicle distribution in isolated neuron: CV (green), DCV (purple), DCVH (orange), SCV (blue), and SDV (red). A highlighted gland cell (G) engulfs the process that has a large number of vesicles. Scale bar, 5 μm (E and F).
See also Figures S4 and S5.

Evolutionary implications

Understanding *Hydra*'s nervous system could be relevant to the structurally similar enteric nervous system of mammals and could also have wider implications for the central or peripheral nervous systems of bilateral species. For example, the interneuronal interdigitations we described may exist in other species, perhaps with a mechanical or gap-junctional role. The classification of neurons using different complements of neuronal vesicle types could also help current efforts to classify the cellular elements of the nervous system in many species and translate ultrastructural databases into cell type information. And, finally, the possibility that at least one part of the nervous system in one animal may function via a non-synaptic mechanism could open the door for the exploration of similar forms of chemical neuroendocrine networks in other species, from non-bilaterians to bilaterians. Similar to synaptic-based neural circuits, chemical networks using neuropeptides may have been widely used by evolution to build neural networks and implement behavioral states.[10]

RESOURCE AVAILABILITY

Lead contact

Requests for further information and resources should be directed to and will be fulfilled by the lead contact, Shulin Zhang (sz2860@columbia.edu).

Materials availability

This study did not generate new, unique reagents.

Data and code availability

VAST[35] segmentation and measurement data have been deposited at Mendeley Data and are publicly available as of the date of publication at https://doi.org/10.17632/5p4ry5pdk2.1.

EM, neuron, and vesicle segmentation data are available in Neuroglancer format at https://doi.org/10.60533/BOSS-2025-08G4.

All original code has been deposited at Zenodo and is publicly available at https://zenodo.org/records/16664362 as of the date of publication.

Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

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AUTHOR CONTRIBUTIONS

S.Z., J.L., and R.Y. conceptualized the work and designed experiments. C.D., X.W., R.S., W.Y., Y.F., and Y.W. prepared and imaged the samples. D.W. designed and supervised computational analysis. S.Z., N.O., M.L., M.R., J.A., A.G., and Y.F. performed reconstructions and analysis. S.Z. and R.Y. wrote the initial manuscript and edited the paper. R.Y. assembled the team, directed the project, and secured resources and funding.

DECLARATION OF INTERESTS

The authors declare no competing interests.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

Supplemental information can be found online at https://doi.org/10.1016/j.cub.2025.10.001.

A video abstract is available at https://doi.org/10.1016/j.cub.2025.10.001#mmc4

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCES	SOURCE	IDENTIFIER
Chemicals		
Osmium Tetroxide 4% Aqueous Solution	Electron Microscopy Sciences	19150; CAS: 20816-12-0
Aqueous Glutaraldehyde EM Grade 25%	Electron Microscopy Sciences	16220; CAS: 111-30-8
TAAB Epon	Marivac Canada Inc.	N/A
Uranyl Acetate, Reagent, A.C.S.	Electron Microscopy Sciences	22400; CAS: 541-09-3
3% Lead Citrate	Leica UltroStain 2	16707235
Deposited data		
Original Dataset	This paper	https://doi.org/10.17632/5p4ry5pdk2.1
EM and segmentation	This paper	https://doi.org/10.60533/BOSS-2025-08G4
Vesicle type visualization	This paper	https://akgohain.github.io/hydra_types/
Vesicle density visualization	This paper	https://akgohain.github.io/hydra_combinedDensity/
Large vesicle density visualization	This paper	https://akgohain.github.io/hydra_lvDensity/
Experimental model		
Hydra vulgaris strain AEP, SS1	Dr. Robert Steele, UC Irvine	N/A
Software and algorithms		
Codes used for the paper	This paper	https://zenodo.org/records/16664362
VAST	Berger et al.[35]	https://lichtman.rc.fas.harvard.edu/vast/
FEABAS	Dr. Yuelong Wu, Harvard University	https://github.com/YuelongWu/feabas
Atlas 5	Zeiss Microscopy	https://www.zeiss.com/microscopy/us/products/software/zeiss-atlas-5.html
Plotly	Plotly	https://plotly.com/python/graph-objects/
three.js	three.js	https://threejs.org
PyVista	The PyVista Project	https://pyvista.org/
scipy.stats	The SciPy community	https://docs.scipy.org/doc/scipy/reference/stats.html
matplotlib.pyplot	The Matplotlib development team	https://docs.scipy.org/doc/scipy/index.html
numpy	NumPy team	https://numpy.org/
Adobe Illustrator	Adobe	N/A

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Hydra

Hydra vulgaris AEP strains were maintained in the dark at 18°C in *Hydra* culture media (1mM CaCl2, 0.5mM NaHCO3, 0.33 mM MgSO4, 0.03 mM KCl). *Hydra* were fed freshly hatched Artemia nauplii three times a week and was starved for 1 week before the fixation.

METHOD DETAILS

SEM acquisition and image alignment

Hydra was placed in the 100 μm deep side of type A 6 mm Cu/Au carriers (Leica), sandwiched with the flat side of type B 6 mm Cu/Au carriers (Leica), and frozen in a high-pressure freezer (EM ICE, Leica). This was followed by freeze-substitution (EM AFS2, Leica) in dry acetone containing 1 % ddH2O, 1 % OsO4, and 0.5 % glutaraldehyde at −90°C for 48 hrs. The temperature was increased 5°C/hr up to 20°C, and samples were washed in pure acetone 3 times, each for 10 minutes, at room temperature and in propylene oxide for

10 minutes. Samples were infiltrated with 1:1 Epon: propylene oxide overnight at 4°C. The samples were subsequently embedded in TAAB Epon (Marivac Canada Inc., St. Laurent, Canada) and polymerized at 60 °C for 48 hrs. The cured specimen blocks were checked for cracks and overall quality using a Zeiss Xradia 510 X-ray microscope. The specimen blocks were scanned using a 50 kV, 4 W beam, with a pixel resolution of 560 nm. The number of steps per 360 degrees was 1201, and the exposure time per step was 40 seconds. Ultrathin sections (30 nm) were cut using a Leica EM UC6 ultramicrotome and collected on continuous carbon-coated Kapton tape using an automated tape collector.[36] Short strips of the Kapton tape holding sections were cut and mounted onto a silicon wafer using double-sided carbon tape (Ted Pella, 16084-8. The sections mounted on wafers were then post-stained for 4 minutes with a 2% uranyl acetate solution, rinsed with DI water, followed by 4-minute staining of 3% lead citrate (Leica UltroStain 2, 16707235), and then rinsed with DI water. Each section was imaged in a Zeiss SIGMA scanning electron microscope using the Atlas 5 (Zeiss Microscopy) mapping software. The imaging parameters included a 1000 ns dwell time, 4 nm pixel size, and back-scattered electron detector imaging. The stitching and alignment of the EM data were performed with the open-source Python feabas.

Manual segmentation

We used VAST[35] for all 20 neurons, 59 volumes, 694,114,926 pixels of large vesicle ground truth and all small vesicle segmentation. The aligned EM data was imported into VAST. Neurons were identified through their distinct morphology, like cilia, the extension of the neurites, vesicles, etc. We then created a labeled segmentation over the identified neuron on all the stacks until it ended. For vesicle ground truth, we created a segmentation label for each type of large vesicle DCV, CV, and DCVH, and segmented the complete vesicles across multiple slices using the corresponding label. For small vesicles, we separated both SCV and SDV with the same label; we did not distinguish their type due to the ambiguous morphology and the possibility of human error.

Machine learning-aided vesicle segmentation and classification

All large vesicles were predicted using a 3D U-NET model trained on ground truth segmentations and proofread to ensure accuracy. The prediction consisted of three channels, including binary segmentation, contour, and distance transform prediction.[37] We generated seeds using contour and distance transform to perform watershed segmentation, which allowed us to separate individual vesicle instances from the binary semantic prediction. We also implemented CLAHE[38] that improved EM image quality, local contrast and prediction accuracy. We performed 4 rounds of iterated prediction and proofreading, then manually corrected the false positive and negative predictions in VAST.

Large vesicle types were classified by a 3D convolutional neural network[39] trained on ground truth labels. Proofread vesicles from the prediction were first processed into 31x31x5 voxel stacks, each containing a single vesicle. The network then processed voxel stacks using a series of convolutional layers with batch normalization and ReLU activations, followed by max pooling and fully connected layers to produce categorical predictions.[39] To mitigate class imbalance and improve generalization, we constructed a balanced training dataset by aggregating equal numbers (1916) of CV, DCV, and DCVH examples. We then created a custom-made proofreading tool to efficiently proofread and correct the label predictions in HTML.

Small vesicles were classified into two morphology types through unsupervised variational autoencoders and clustering. We first extracted image patches centered around each vesicle and resized and standardized them into 11x11 pixel patches. We then utilized variational autoencoders with in-built translation and rotation invariance, provided by the pyroVED[40] library. Using this model, each image patch was compressed into an embedding with two latent dimensions. Finally, we used the HDBSCAN algorithm[41] to cluster the embedding into two categories (determined by visual inspection). Detailed computational pipeline. [42]

Spatial, morphology and density analysis

Vesicle categorizations of "near" or "far" from other neurons were defined by a Euclidean Distance Transform to the neuronal membranes of any adjacent neurons—classifying "near" and "far" regions based on distance thresholds determined by the average sizes of all vesicles of the same type. Neurons and vesicles were stored as binary masks in HDF5 format, each containing the smallest bounding box region around a single neuron. We computed volumes of each vesicle through a vectorized summation of voxels, and computed radii by collapsing each vesicle into two dimensions and extracting the area of the resultant roughly circular region. The dataset of vesicles was then stored in point cloud metadata format: as a list of the coordinates of the centers of mass of each vesicle, together with corresponding information including volume, radius, segmentation ID, type, and density in terms of the presence of nearby vesicles. We then performed statistical analysis, using the vesicle diameters to determine the "near" and "far" thresholds for each vesicle type. To factor in adjacent neurons for our calculations, we used a global coordinate system to locate and stitch together any pieces of other neurons within the bounding box of each target neuron. The relevant pieces of adjacent neurons were then expanded by previously calculated distance thresholds for each vesicle type, and the vesicles overlapping with the expanded area were classified as "near" while vesicles outside of the region were "far". Finally, we computed the overall spatial distribution of vesicles by using the point cloud to construct a kdtree for each neuron, finding the relative amount of nearby vesicles within 500nm of each vesicle to demonstrate which regions are more "dense" in terms of the presence of vesicles. Detailed computational pipeline.[42]

Data visualization

We deployed Neuroglancer for web-based visualization of neuron and vesicle segmentations. Segmentations were converted into Neuroglancer-readable precomputed format with accompanying mesh and skeleton layers. Vesicles were exported as point clouds (with ID, type, volume, and coordinates) and displayed as annotation layers. Point cloud annotations could be visualized either

colored by vesicle subtype or by vesicle density, calculated using a 500 nm-radius kdtree-based neighborhood metric. We used PyVista[43] to render interactive 3D meshes of neurons and vesicles locally. Vesicle segmentations were converted into .obj format and visualized alongside transparent neuron meshes. We mapped vesicle types to color codes and overlaid density values using scalar fields. Plotly was used for lightweight browser-friendly 3D visualizations of vesicle center-of-mass point clouds. Coordinates and metadata were stored in .parquet and plotted as scatter plots with subtype, size, and density attributes mapped to color and size. A custom three.js viewer was implemented for dynamic 3D exploration of large-scale vesicle point clouds. Vesicle attributes were visualized interactively using bounding boxes and density-based color gradients. Detailed computational pipeline.[42]

Extended data 1

Interactive HTML of spatially averaged vesicles color coded by type CV (green), DV (purple), DCVH (orange), SCV (blue), SDV (red) in complete neurons. https://akgohain.github.io/hydra_types/. related to Figure 7

Extended data 2

Interactive HTML of spatially averaged vesicles color coded by density (number of vesicles within 0.5 μm^3, 0:blue, 15+:red) in complete neurons. https://akgohain.github.io/hydra_combinedDensity/. related to Figure 7

Extended data 3

Interactive HTML of spatially averaged vesicles, excluding small vesicles color-coded by density (number of vesicles within 0.5 μm^3, 0:blue, 15+:red) in complete neurons. https://akgohain.github.io/hydra_lvDensity/. related to Figure 7

QUANTIFICATION AND STATISTICAL ANALYSIS

Measurements and counts

The size of the animal, cilia, microvilli, bulbous structure, handshake structure and neuron length were measured in VAST using the skeleton tool and exported by length. The hypostome volume was quantified using VAST segmentation tool annotating over the hypostome and body column region (excluding tentacle and gastric cavity) on the middle slice of the sample, then exported by volume. The nucleus volume was also segmented manually and exported using VAST. Fusing and docking vesicles were counted manually in VAST.

Statistics

All statistics were computed in Python using scipy.stats and in Google Sheets. Graphs were created in Python using matplotlib.pyplot and numpy, and Adobe Illustrator was used for formatting.